## VECTORS

| *Term* | *Definition* |
|---|---|
| Norm/Length | $\lVert v\rVert = \sqrt{\sum_{i=1}^{n} v_i^2}$ <br> $\lVert v\rVert = \sqrt{v^T v}$ <br> $\nabla\lVert v\rVert = \frac{1}{\lVert v\rVert} v^T$ |
| Distance | $d(u,v) = \lVert u - v\rVert$ <br> **TODO:** <br> $\nabla d(u,v) = \frac{1}{d(u,v)}((u-v)(u-v))???$ <br> $\nabla_u d(u,v) = \frac{u-v}{\lVert u-v\rVert} = \frac{u-v}{d(u,v)}$ <br> $\nabla_v d(u,v) = -\frac{u-v}{\lVert u-v\rVert} = \frac{v-u}{d(u,v)}$ |
| Scalar product | $u \bullet v = u^T v = \sum_k u_k v_k$ |

## SCALAR PRODUCT

## MATRICES

**Determinant**:

$$\det(a) = a \qquad \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

**Inverting**:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

**Transposing**:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1\ 4\ 5)$$

**Matrix multiplication**:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$

$$= \begin{pmatrix} (2\cdot 6 + 3\cdot 1 + 1\cdot 8) & (2\cdot 4 + 3\cdot 0 + 1\cdot 9) \\ (4\cdot 6 + -1\cdot 1 + 7\cdot 8) & (4\cdot 4 + -1\cdot 0 + 7\cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

## INDICES

$$A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{m\times r}, A\cdot B \in \mathbb{R}^{n\times r}$$

$$(A\cdot B)_{ij} = \sum_{k=1}^{m} A_{ik}B_{kj} = \sum_k A_{ik}B_{kj}$$

$$((A\cdot B)^T)_{ij} = (A\cdot B)_{ji} = \sum_k A_{jk}B_{ik}$$

$$a^T \cdot B \cdot a = \sum_{ij} (a^T)_i B_{ij} a_j = \sum_{ij} a_i B_{ij} a_j$$

$$(B\cdot a)(C\cdot a) = \sum_i (B\cdot a)_i (C\cdot a)_i$$

## KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1 & , i = j \\ 0 & , \text{otherwise} \end{cases}$$

$$(E_{rst})_{ijk} = \delta_{ri}\delta_{sj}\delta_{tk}$$

## BASIS VECTORS

$\mathbb{R}^3$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

## UNIT MATRICES

$\mathbb{R}^2$

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

## AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M\cdot x$$

$$L_A(\vec{0}) = \vec{0}$$

## DERIVATIVES

| | | | |
|---|---|---|---|
| $1$ | $\to 0$ | $x$ | $\to 1$ |
| $x^2$ | $\to 2x$ | $x^n$ | $\to nx^{n-1}$ |
| $\frac{1}{x}$ | $\to -\frac{1}{x^2}$ | $\sqrt{x}$ | $\to \frac{1}{2\sqrt{x}}$ |
| $e^x$ | $\to e^x$ | $e^{-x}$ | $\to -e^{-x}$ |
| $\ln(x)$ | $\to \frac{1}{x}$ for $x > 0$ | $\ln(y\cdot x)$ | $\to \frac{1}{x}$ for $x > 0$ |
| $a^x$ | $\to \ln(a)\cdot a^x$ for $a > 0, a \neq 1$ | $\log_b(x)$ | $\to \frac{1}{\ln(b)\cdot x}$ |
| $\sin(x)$ | $\to \cos(x)$ | $\sin(2x)$ | $\to 2\cos(2x)$ |
| $\cos(x)$ | $\to -\sin(x)$ | $\cos(ax)$ | $\to -a\sin(ax)$ |
| $\tan(x)$ | $\to 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$ | $\operatorname{arccot}(x)$ | $\to -\frac{1}{1+x^2}$ |
| $\cot(x)$ | $\to -\frac{1}{\sin^2(x)}$ | $\arcsin(x)$ | $\to \frac{1}{\sqrt{1-x^2}}$ |
| $\arccos(x)$ | $\to -\frac{1}{\sqrt{1-x^2}}$ | $\arctan(x)$ | $\to \frac{1}{1+x^2}$ |

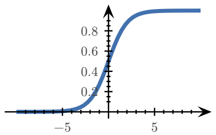
| *Term* | *Definition* |
|---|---|
| Addition | $(f(x) + g(x))' = f'(x) + g'(x)$ |
| Multiplication | $(\alpha\cdot f(x))' = \alpha\cdot f'(x)$ |
| Product rule | $(f\cdot g)' = f'\cdot g + f\cdot g'$ <br> $(f\cdot g\cdot h)' = f'\cdot g\cdot h + f\cdot g'\cdot h + f\cdot g\cdot h'$ |
| Chain rule | $(f(g(x)))' = f'(g(x))\cdot g'(x)$ |
| Quotient rule | $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ |

## FUNCTIONS OF SEVERAL VARIABLES

| *Term* | *Definition* |
|---|---|
| Real valued function | $R \subset \mathbb{R}$ |
| Vector valued function | $R \subset \mathbb{R}^m$ |
| A function of $n$ variables | $D \subset \mathbb{R}^n$ |
| Softmax function | **TODO:** |
| RSS | $\text{RSS} = \sum_i (y_i - f(x_i))^2$ |
| $n$-dimensional curve | $c : \mathbb{R} \to \mathbb{R}^n$ |
| $n$-dimensional hyper-surface | $s : \mathbb{R}^n \to \mathbb{R}$ |
| Reparametrization | $c(t) \to d(s) = c(h(s))$ |

**Sigmoid function**:

$$\sigma(x) : \mathbb{R} \to [0;1] = \frac{1}{1+e^{-x}}$$




## DERIVATIVES

| *Function* | *Derivative* |
|---|---|
| $f : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{\mathrm{d}}{\mathrm{d}x} f_1(x) \\ \frac{\mathrm{d}}{\mathrm{d}x} f_2(x) \\ \vdots \\ \frac{\mathrm{d}}{\mathrm{d}x} f_n(x) \end{pmatrix}$ |
| $f : \begin{cases} \mathbb{R}^n \to \mathbb{R} \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{1\times n} \\ x \mapsto y \end{cases} = \nabla f$ <br> $= \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), ..., \frac{\partial}{\partial x_n} f(x)\right)$ |
| $f : \begin{cases} \mathbb{R}^n \to \mathbb{R}^m \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{m\times n} \\ x \mapsto y \end{cases} = J_f = \left.\frac{\partial f}{\partial x}\right\vert_{x=x_0}$ <br> $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix}$ |

### PARTIAL VS TOTAL DERIVATIVE

**TODO:**
example FS2024

$$f(x,y) = x^2 + 2y$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 2\cdot\frac{\mathrm{d}y}{\mathrm{d}x} \mid (\text{y dependent on x})$$

### GRADIENTS

**Common gradients**:

$$f(x) = \lVert x\rVert^2 \Rightarrow \nabla f = 2x$$

$$f(x) = a^T x + c \Rightarrow \nabla f = a$$

$$f(x) = x^T A x \Rightarrow \nabla f = (A + A^T)x$$

$$f(x) = \frac{1}{2}x^T A x + b^T x + c \Rightarrow \nabla f = \frac{A + A^T}{2}x + b$$

$$= Ax + b \text{ if } A^T = A$$

$$f(x) = g(Ax + b) \Rightarrow \nabla f = A^T \nabla g(Ax + b)$$

$$g : \mathbb{R}^m \to \mathbb{R}$$

$$f(x) = u(x)\cdot v(x) \Rightarrow \nabla f = (J_u)^T v + (J_v)^T u$$

$$f(x) = \lVert u(x)\rVert = \sqrt{u^T u} \Rightarrow \nabla f = (J_u)^T \frac{u}{\lVert u\rVert} \text{ if } \lVert u\rVert \neq 0$$

$$f(x) = \log p(x) \Rightarrow \nabla f = \frac{1}{p(x)}\nabla p(x)$$

### JACOBIAN MATRIX

$$f : \mathbb{R}^2 \to \mathbb{R}^2,\ J_f(x,y) = \begin{pmatrix} \partial x_1 & \partial y_1 \\ \partial x_2 & \partial y_2 \end{pmatrix}$$

**Linearisation**:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ x_0 \in \mathbb{R}^n$$

$$L_f(x) = f(x_0) + J_f(x_0)\cdot(x - x_0)$$

**Chain rule**:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R}^k,\ h : \mathbb{R}^n \to \mathbb{R}^k = g\circ f$$

$$J_h = J_g(f(x))\cdot J_f(x)$$

**Common jacobians**:

$$f(x) = a + M\cdot x \Rightarrow J_f = M$$

$$f(x) = x \Rightarrow J_f = \mathbb{1}$$

$$f(x) = \lVert x\rVert^2 \Rightarrow J_f = (2x)^T$$

**TODO:**
$J_x$ for softmax $s_i = \frac{e^{x_i}}{\Sigma}e^{x_k} : J_{ij} = s_{i(\delta_{ij} - s_j)}$

### HYPER-SURFACES

**Chain rule**:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R},\ h : \mathbb{R}^n \to \mathbb{R} = g\circ f$$

$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f)\vert_{f=f(x)}\cdot\frac{\partial}{\partial x}f(x)$$

**Differentiation properties**:

$$f : \mathbb{R}^n \to \mathbb{R},\ g : \mathbb{R}^n \to \mathbb{R}$$

$$\nabla(f + g) = \nabla f + \nabla g$$

$$\nabla(f - g) = \nabla f - \nabla g$$

$$\nabla(f\cdot g) = g\nabla f + f\nabla g$$

$$\nabla\frac{f}{g} = \frac{g\nabla f - f\nabla g}{g^2}$$

## COMMUTATIVE DIAGRAMS

$f : D \to M$

$g : N \to R$

$h : D \to R$

$= g\circ f$

$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] & M \arrow[r, "g"] & R \end{tikzcd}$$

## GEOMETRY OF HYPER-SURFACES

$$f : \mathbb{R}^2 \to \mathbb{R}$$

**Tangent space**: $\alpha, \beta \in \mathbb{R}, T =$

$$\left\{\begin{pmatrix} x_0 \\ y_0 \\ f(x_0, y_0) \end{pmatrix} + \alpha\begin{pmatrix} 1 \\ 0 \\ \partial_x f(x_0, y_0) \end{pmatrix} + \beta\begin{pmatrix} 0 \\ 1 \\ \partial_y f(x_0, y_0) \end{pmatrix}\right\}$$

**Graph of a linearisation**:

$$l(x) = f(x_0) + \nabla f(x_0)\cdot(x - x_0) = T_{p_0}$$

### SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right) = \frac{\partial^2 f(x,y)}{\partial x^2} = \partial_{xx} f(x,y)$$

### HESSIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}$$

$$H(x,y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$

$$(H(x))_{ij} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f(x)$$

$$\partial_i\partial_j f = \partial_j\partial_i f$$

**Common hessians**:

$$f(x) = \frac{1}{2}x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$$

$$f(x) = g(Ax + b) \Rightarrow H_f = A^T H_g(Ax + b)A$$

$$g : \mathbb{R}^m \to \mathbb{R}$$

### EXTREMAL POINTS

Let $z = f(c(t)),\ c$ a curve passing the stationary point $c_0 = (x_0, y_0)$ at $t = t_0$ and

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T}\cdot\underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0) \end{pmatrix}}_{H}\cdot\underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0) \end{pmatrix}}_{v}$$

1) If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\}(v^T H v > 0)$, then $\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} > 0$ defines a ***local minimum***
2) If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\}(v^T H v < 0)$, then $\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} < 0$ defines a ***local maximum***
3) If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\}(v^T H v < 0 \vee v^T H v > 0)$, the point $c_0 = (x_0, y_0)$ defines a ***saddle point***

**TODO:**

$$(J_u)_{ij} = \frac{\partial u_i}{\partial x_j}, (H_g)_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$$

$$\partial_j(u_i v_i) = (\partial_j u_i)v_i + u_i(\partial_j v_i)$$

## MISC

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## TRIGONOMETRY

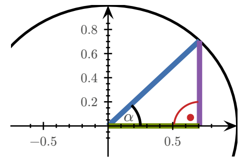
| $x$ | $0$ | $30° = \frac{\pi}{6}$ | $45° = \frac{\pi}{4}$ | $60° = \frac{\pi}{3}$ | $90° = \frac{\pi}{2}$ |
|---|---|---|---|---|---|
| $\sin(x)$ | $0$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $1$ |
| $\cos(x)$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ |
| $\tan(x)$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | |

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$\sin^2(x) + \cos^2(x) = 1$$

$1, \sin(\alpha), \cos(\alpha)$




## GRADIENT DESCENT / NEWTON'S METHOD

$$GD : x_{i+1} = x_i - \gamma\cdot\nabla f(x_i)$$

$$N : x_{i+1} = x_i - \gamma\cdot\left(H_{f(x_i)}\right)^{-1}\cdot\nabla f(x_i)$$

$$\gamma = \text{step size}$$

$$\text{termination criterion } \varepsilon > \lvert x_{i+1} - x_i\rvert$$

## EXAMPLES

**TODO:**